DEPARTMENT OF AGRICULTURE IV-A · DEPDev IV-A

Food Insecurity Risk Forecast

Laguna · Q4 2025 (Oct-Dec)

Generated 29 September 2026 from aiPHeed model output.

What this number is

The modelled probability that this area falls in the HIGH class of the composite food stress label. **It is an indicative relative risk estimate, not an official food insecurity classification**, and it is published at the Maximum Reliable Lead Time the model established empirically.

Headline

FOOD INSECURITY RISK PROBABILITY

0.528

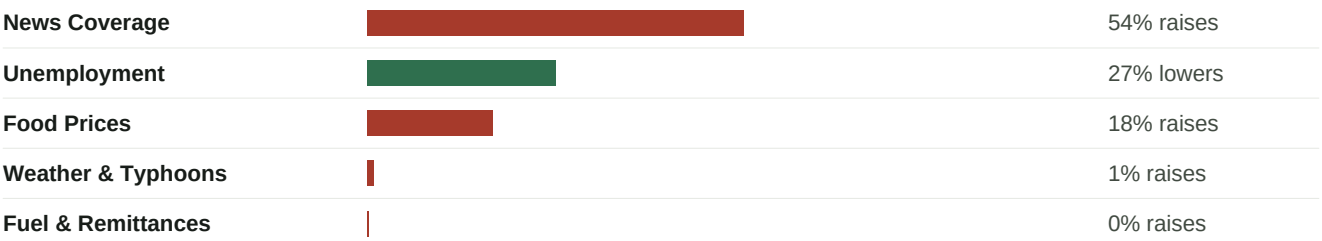
HIGH · quarter-on-quarter +0.0000

Risk class	**HIGH**
Rank in CALABARZON	**1 of 5**
Population (2020 CPH)	**3,382,193**
Poverty incidence (PSA 2021)	**4.2%**
Articles analysed	**0**

Forecast horizon

Same-quarter estimate. No forward lead time met the 75% reliability threshold of Section 3.4.4, so none is published.. Indicative relative risk only. Walk-forward accuracy did not reach 75% at any forward horizon, so this output is reported as a same-quarter estimate and not as a forecast.

What is driving the figure



Red bars raise the predicted shortfall, **green** lower it. A large bar is not automatically bad news -- share of the explanation and direction of effect are separate things.

For Laguna in 2025-Q4, the model's prediction breaks down as: News Coverage 54%, Unemployment 27%, Food Prices 18%, Weather & Typhoons 1%, Fuel & Remittances 0%. News Coverage is the largest single contributor, pushing the score up. News Coverage clears the 20% line while raising the score. Unemployment clears the 20% line while lowering it. Shares are SHAP contributions to the province's food insecurity risk probability, recomputed every quarter.

Municipal distribution

Municipal values reweight the province figure by poverty incidence (60%) and population density (40%). **They are not independent municipal forecasts** -- every municipality in a province rises and falls with it.

#	City / Municipality	Class	Population	Poverty	Index
1	**San Pablo**	City	285,348	6.4%	0.3376
2	**San Pedro**	City	326,001	4.2%	0.3288
3	**Biñan**	City	407,437	4.2%	0.2748
4	**Cabuyao**	City	355,330	4.2%	0.2448
5	**Santa Cruz**	Municipality	123,574	4.2%	0.1661
6	**Los Baños**	Municipality	115,353	4.2%	0.1492
7	**Victoria**	Municipality	43,408	4.2%	0.1463
8	**Pila**	Municipality	54,613	4.2%	0.1433
9	**Pagsanjan**	Municipality	44,327	4.2%	0.1412
10	**Bay**	Municipality	67,182	4.2%	0.1405
11	**Calamba**	City	539,671	3.8%	0.1363
12	**Calauan**	Municipality	87,693	4.2%	0.1335
13	**Liliw**	Municipality	39,491	4.2%	0.1317
14	**Alaminos**	Municipality	51,619	4.2%	0.1299
15	**Nagcarlan**	Municipality	64,866	4.2%	0.1288
16	**Lumban**	Municipality	32,330	4.2%	0.1283
17	**Magdalena**	Municipality	27,816	4.2%	0.1283
18	**Rizal**	Municipality	18,332	4.2%	0.1261
19	**Siniloan**	Municipality	39,460	4.2%	0.1254
20	**Pangil**	Municipality	25,026	4.2%	0.1245
21	**Kalayaan**	Municipality	24,755	4.2%	0.1241

#	City / Municipality	Class	Population	Poverty	Index
22	**Pakil**	Municipality	23,495	4.2%	0.1237
23	**Paete**	Municipality	24,945	4.2%	0.1229
24	**Majayjay**	Municipality	27,893	4.2%	0.1221
25	**Famy**	Municipality	16,791	4.2%	0.1208
26	**Santa Maria**	Municipality	34,511	4.2%	0.1208
27	**Luisiana**	Municipality	20,859	4.2%	0.1203
28	**Mabitac**	Municipality	21,275	4.2%	0.1199
29	**Cavinti**	Municipality	23,980	4.2%	0.1176
30	**Santa Rosa**	City	414,812	2.9%	0.1170

News evidence

No relevant articles were geocoded to this subject for the quarter.

Method

The model predicts, for each CALABARZON province, the probability that the province falls in the HIGH class of the composite food stress label -- SWS quarterly hunger prevalence combined with the province's food CPI deviation from the regional mean, thresholded at the global median. Inputs are the bias-corrected Food Stress Sentiment Index and its lags and acceleration, the five bilingual trigger proportions, and government indicators from PSA, BSP, DOE, PAGASA and PhilRice.

Measure	Value
Maximum Reliable Lead Time	0 quarter(s)
Accuracy	0.6000
Precision (weighted)	0.9200
Recall (weighted)	0.6000
F1 score (weighted)	0.6762
AUC-ROC	0.9444
F1 target (>= 0.75)	FAIL
AUC-ROC target (>= 0.80)	PASS
Features in the model	12
Baseline: aipheed lgbm combined	0.6727
Baseline: full feature logistic	0.5455
Baseline: lgbm nlp only	0.6182
Baseline: lgbm psa only	0.7273
Baseline: naive persistence	0.8000
Baseline: nlp only logistic	0.5455
Baseline: psa only logistic	0.5818

Accuracy must be read against its baselines. The comparison models above are evaluated on the same walk-forward folds at the same forecast gap, so the figures are directly comparable. Naive persistence in particular is a demanding baseline on an autocorrelated label: a model that does not clearly beat it is not earning its complexity.

Limitations

Indicative risk, not official classification

Outputs are relative ordinal estimates of potential risk. They are not definitive determinations of food insecurity status and must not be read as official government assessments or policy recommendations.

Proxy label, not a direct measurement

No publicly accessible province-quarter food insecurity label exists in the Philippines. The training target is a composite of SWS hunger prevalence, which has no province-level resolution, and PSA food CPI deviation, which reflects supply-side price conditions rather than household hunger experience directly.

Municipal figures are disaggregated, not modelled

Municipal values reweight the province figure by poverty incidence (60%) and population density (40%). Poverty values are largely province means rather than municipal Small Area Estimates, so within-province ordering is indicative only.

Reviewed before publication

Province-quarter figures pass through analyst review. A withheld figure is excluded from this report and from the regional average.

Scope of this assessment

This assessment reports findings only. It does not recommend interventions: no validated mapping exists between this indicator and a response protocol. Pair it with the Department's own standing guidance.